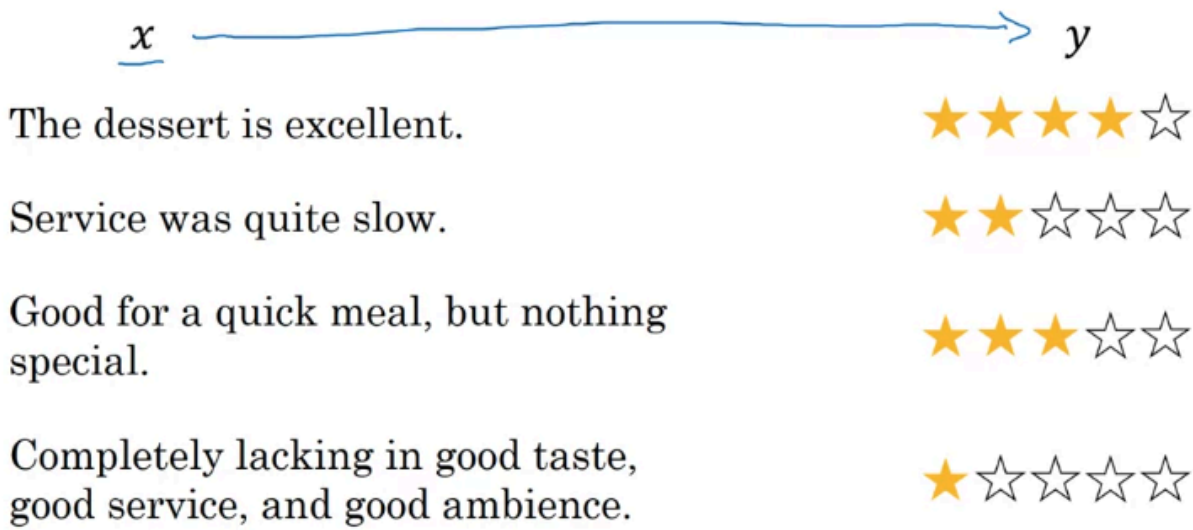


Applications Using Word Embeddings

Sentiment Classification

Given comments and predict its ratings

Sentiment classification problem

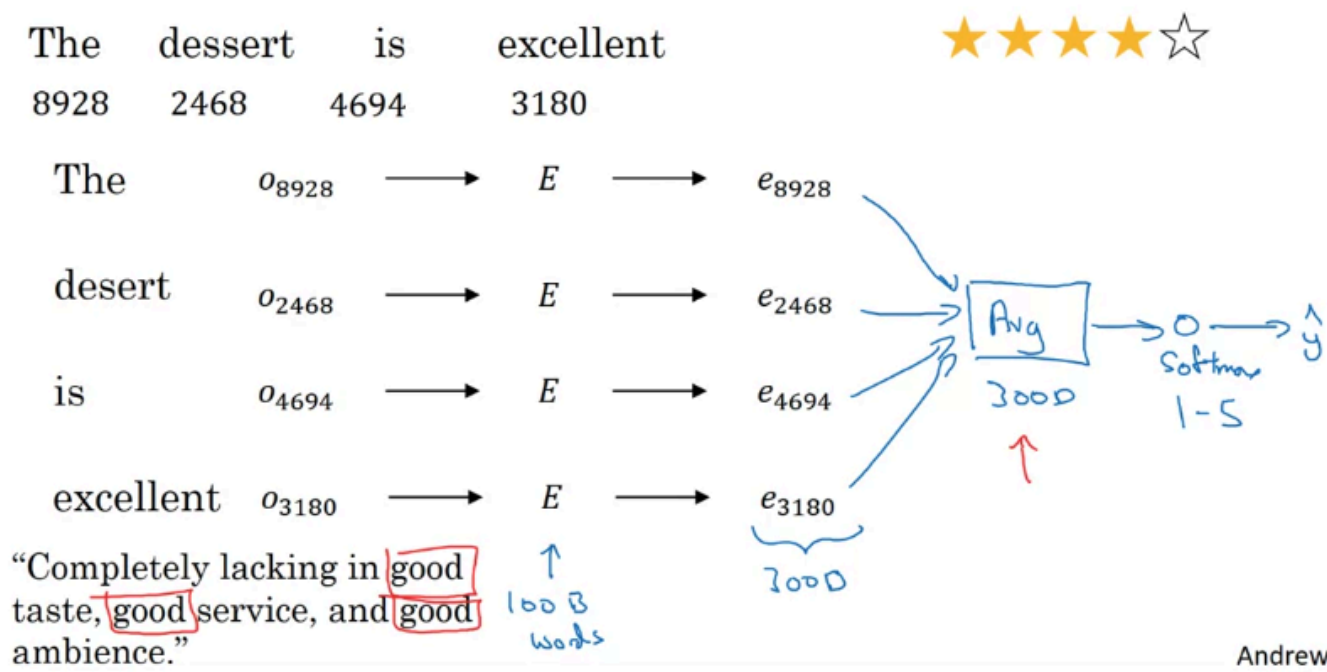


So for sentimental classification task,

Andrew Ng

Simple sentiment classification model

Simple sentiment classification model



Andrew Ng

Get the embeddings of the input words, and fusion it (eg: average, sum, ...) then put the fused vector into the softmax and get the predicted rating

Problem: Word Order Is Ignored

A major weakness of averaging is that it treats the sentence roughly as a **bag of words**.

Consider:

Completely lacking in good taste, good service, and good ambiance.

The word **good** appears several times.

A simple averaging model may therefore see many positive embedding signals and mistakenly classify the review as positive.

However, the actual meaning is strongly negative because of phrases such as:

lacking in good taste

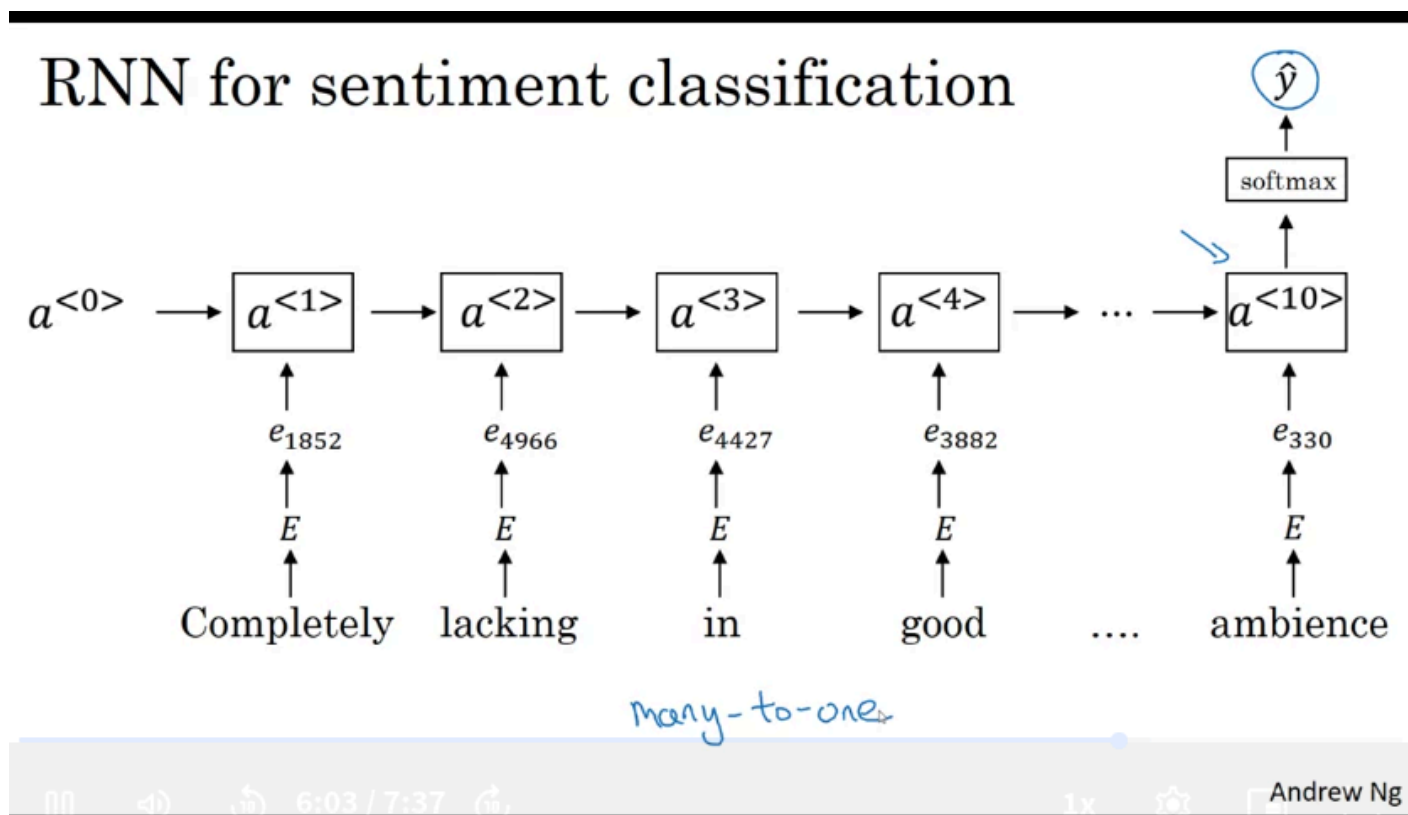
So:

meaning $\neq$ just average of individual words

The problem:

- If the comment has many "good" words but its meaning is bad (like the example above) the model will be affected by the word "good" in the fusion step $\rightarrow$ the output rating could be wrong

RNN for sentiment classification



Transfer Learning from Pretrained Embeddings

One major advantage comes from using embeddings trained on a huge corpus.

Suppose the sentiment training set contains:

lacking in good taste

but never contains the word:

absent

A new review might say:

Completely absent of good taste.

If the pretrained embeddings learned that:

$$e_{\text{absent}} \approx e_{\text{lacking}}$$

then the sentiment classifier may still generalize correctly even though **absent** never appeared in the labeled sentiment dataset.

Debiasing Word Embeddings

The problem of bias in word embeddings

Man:Woman as King:Queen

Man:Computer_Programmer as Woman:Homemaker ✗

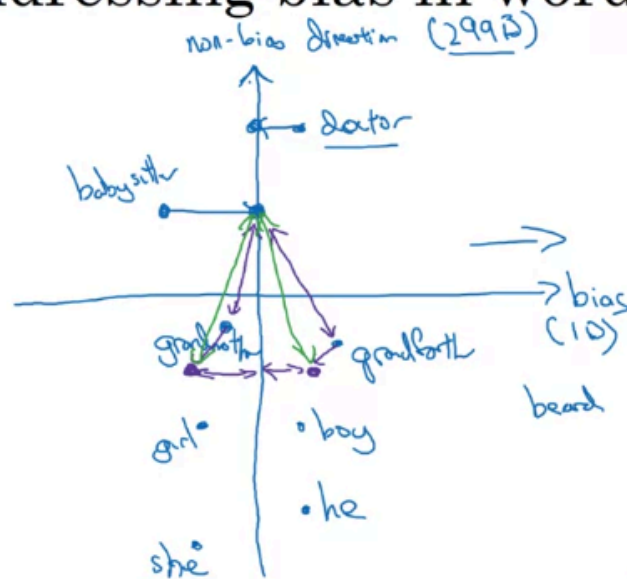
Father:Doctor as Mother:Nurse ✗

Word embeddings can reflect gender, ethnicity, age, sexual orientation, and other biases of the text used to train the model.

sexual orientation, and other biases
of the text used to train the model

[Bolukbasi et. al., 2016. Man is to computer programmer as woman is to homemaker? Debiasing word embeddings] Andrew Ng

Addressing bias in word embeddings



1. Identify bias direction.

$$\begin{cases} e_{\text{he}} - e_{\text{she}} \\ e_{\text{male}} - e_{\text{female}} \\ \vdots \\ \text{average} \end{cases}$$

2. Neutralize: For every word that is not definitional, project to get rid of bias.

3. Equalize pairs.

$$\rightarrow \begin{cases} \text{grandmother} - \text{grandfather} \\ \text{girl} - \text{boy} \end{cases}$$

[Bolukbasi et. al., 2016. Man is to computer programmer as woman is to homemaker? Debiasing word embeddings] Andrew Ng

Bias in Word Embeddings

- Word embeddings can unintentionally learn **undesirable social biases** from the text used to train them.
 - These may include:
 - Gender bias
 - Ethnicity bias
 - Sexual orientation bias
 - Age bias
 - Socioeconomic bias
 - This is especially important because machine learning systems are increasingly used in high-impact decisions.
-

Example of Undesirable Bias

Word embeddings may learn analogies such as:

man : computer programmer $\approx$ woman : homemaker

or:

father : doctor $\approx$ mother : nurse

These relationships may reflect stereotypes present in the training corpus rather than desirable or fair associations.

Why Bias Appears

- Word embeddings learn statistical patterns from large text corpora.
 - If the training data contains biased associations, the embedding space can reproduce them.
 - Therefore, embeddings may reflect historical and social biases present in human-written text.
-

Step 1: Identify the Bias Direction

The first step is to find a direction in embedding space corresponding to the bias of interest.

For gender, differences such as: $e_{\text{he}} - e_{\text{she}}$

or: $e_{\text{male}} - e_{\text{female}}$

can help identify a **gender direction**.

Multiple gender-related difference vectors can be combined to estimate this direction.

In the simplified example:

- Bias direction: approximately **1-dimensional**
 - Non-bias subspace: approximately **299-dimensional** for 300-D embeddings
-

Finding the Bias Subspace

- The simple intuition is to average several gender-difference vectors.
 - In the actual method, a more sophisticated technique such as **Singular Value Decomposition (SVD)** is used.
 - This is related to ideas from **Principal Component Analysis (PCA)**.
 - The bias subspace can also contain **more than one dimension**.
-

Step 2: Neutralization

Some words should ideally be neutral with respect to gender.

Examples:

- doctor

- babysitter

For these words, remove their component along the bias direction.

Conceptually: $e_{\text{neutral}} = e - \text{projection of } e \text{ onto bias direction}$

This projects the word into the **non-bias subspace**.

Definitional vs. Non-Definitional Words

Not every word should be neutralized.

Definitional Words

Gender is inherently part of their meaning:

- grandmother / grandfather
- girl / boy
- she / he
- sister / brother
- daughter / son

These words should retain their gender information.

Non-Definitional Words

Gender should generally not be part of their meaning:

- doctor
- programmer
- babysitter

These are candidates for neutralization.

Step 3: Equalization

Some gender-specific word pairs should differ **only in gender**.

Examples:

- grandmother / grandfather
- girl / boy
- sister / brother
- niece / nephew
- daughter / son

The goal is to make the two words equally related to gender-neutral concepts.

For example:

$\text{similarity}(\text{babysitter}, \text{grandmother})$

should ideally equal:

$\text{similarity}(\text{babysitter}, \text{grandfather})$

if **babysitter** is intended to be gender-neutral.

Intuition Behind Equalization

Before equalization:

$$d(\text{babysitter}, \text{grandmother}) \neq d(\text{babysitter}, \text{grandfather})$$

After equalization:

$$d(\text{babysitter}, \text{grandmother}) = d(\text{babysitter}, \text{grandfather})$$

The pair is repositioned symmetrically with respect to the neutral subspace.

Choosing Which Words to Neutralize

- Most words are **not definitional** with respect to gender.
 - A classifier can be trained to determine whether a word should:
 - Keep its gender component, or
 - Be neutralized
 - The set of definitional words is relatively small compared with the whole vocabulary.
-

Some Associations May Be Legitimate

Not every statistical gender association should necessarily be removed.

For example:

| beard

may statistically occur more often in contexts associated with men.

Therefore, deciding what counts as undesirable bias is not always trivial.

Debiasing Procedure

The overall procedure can be summarized as:

Identify Bias Direction → Neutralize → Equalize

1. Identify

Find the subspace representing the unwanted bias.

2. Neutralize

Remove the bias component from words that should be neutral.

3. Equalize

Make appropriate definitional word pairs symmetric relative to the neutral subspace.

Key Takeaways

- Word embeddings can inherit social biases from their training data.
- Bias can appear as undesirable semantic relationships in embedding space.
- One debiasing approach consists of:
 1. Finding a **bias direction/subspace**
 2. **Neutralizing** non-definitional words
 3. **Equalizing** appropriate word pairs
- Some words legitimately contain gender information, so they should not be neutralized.
- Debiasing machine learning systems remains an **active research area**, especially because these systems are increasingly used in important real-world decisions.